Lecture 22 DAA April 8

— Hashing —

You have an universe $\mathcal{U}$ of all possible keys, say $0, 1, 2, \ldots N-1$

$$|\mathcal{U}| = N$$

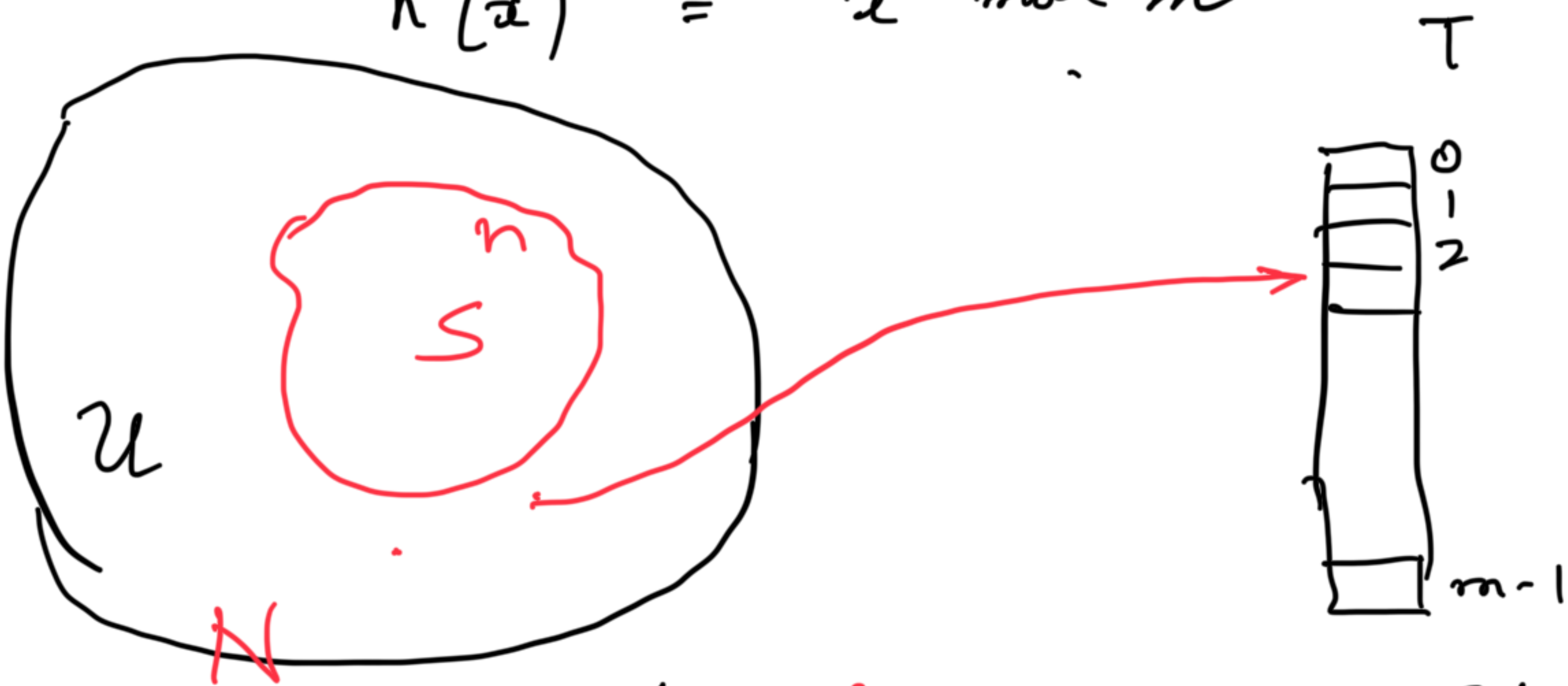
A specific subset $S \subset \mathcal{U}$ represents the set of n keys that will be stored in a hash table T of size m

$$h : \mathcal{U} \rightarrow T$$

$$\{0 .. N-1\} \rightarrow \{0 .. m-1\}$$

A commonly ~~based~~ hash function

$$h(x) = x \bmod m$$



Collision / Conflict $\quad x \neq y \in \mathcal{U}$

$$h(x) = h(y)$$

Resolving collision

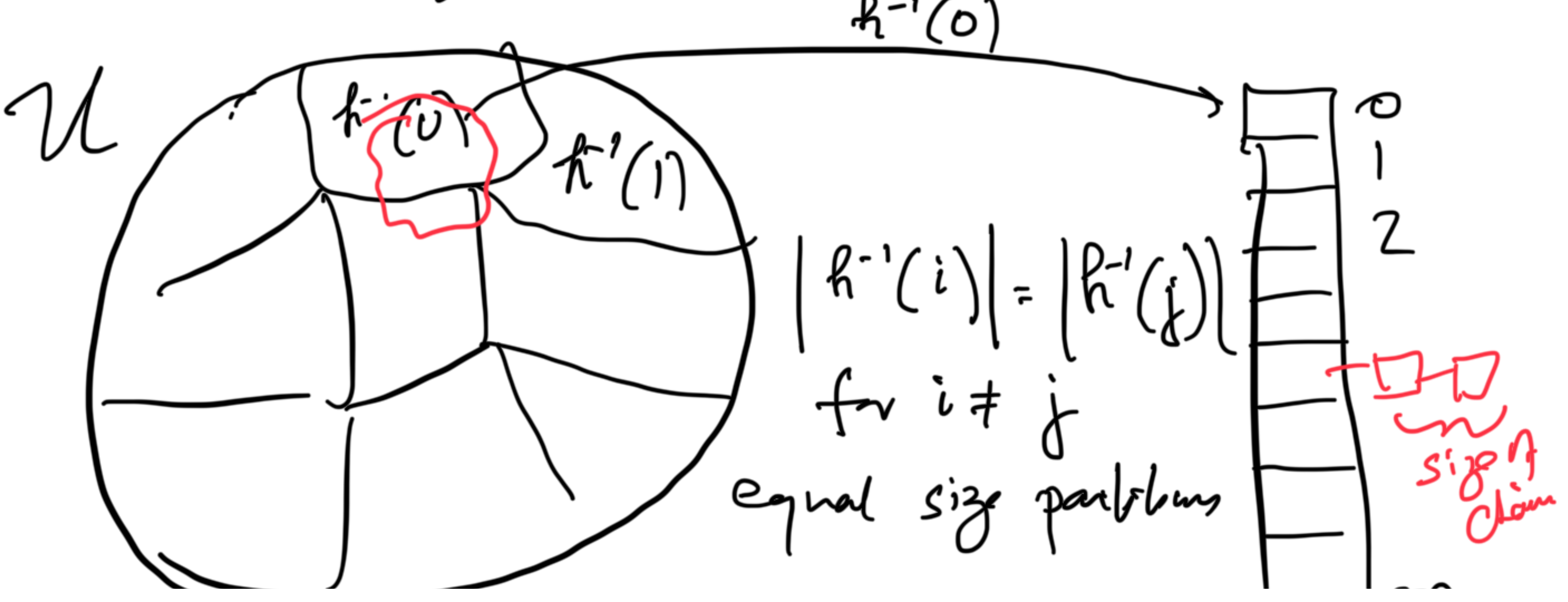
(1) Open addressing : find the next available unoccupied location

✓ (2) Chaining ...

Chaining : Create a link list with the location

Observation : For bad inputs S, there can be a large # of collisions

$\Rightarrow$ much more than $O(1)$ time perhaps even $\Omega(n)$ if all keys get hashed to the same location



Assume that the keys x are picked from $\mathcal{U}$ with uniform distribution independently

Prob ($x \in S$) gets hashed to location i, $0 \le i \le m-1$

$= \frac{1}{m}$

Expected # of elements hashed to location 0

$\frac{|S|}{m} = \frac{n}{m}$ (sum of expectation)

$= O(1)$ when $n \sim m$

$\Rightarrow$ The expected length of chain is $O(1)$

$\Rightarrow$ The expected search / insertion / deletion is $O(1)$

Let H denote a family of hash functions and each function $h \in H$

$$h : \mathcal{U} \to T$$

We will choose a function uniformly at random that will remain unchanged for S.

Let $\delta_h(x,y) = \begin{cases} 1 & \text{if } h(x) = h(y) \\ 0 & \text{otherwise} \end{cases}$

collision function

$$\delta_h(x, S) = \sum_{\substack{y \in S \\ y \neq x}} \delta_h(x, y)$$

= total # of collisions of x with the remaining elements of S

H is called c-universal if

$$\forall\, x, y \in \mathcal{U} \quad x \neq y$$

$$\sum_{h \in H} \delta_h(x, y) \leq c \cdot \frac{|H|}{m}$$

$\Rightarrow$ 1. The fraction of hash function for which any pair collides is $\sim \frac{1}{|T|}$

2. If we pick a hash function at

random, the probability of collision of x, y is $\sim \frac{c}{m}$

$$\frac{1}{|H|} \sum_{\substack{h \in H \\ x \in S}} \delta_h(x, S) = \frac{1}{|H|} \sum_{h \in H} \sum_{y \in S} \delta_h(x, y)$$

$$= \frac{1}{|H|} \sum_{y \in S} \sum_{h \in H} \delta_h(x, y)$$

$$= |S| \cdot \left(\frac{1}{|H|} \sum_{h \in H} \delta_h(x, y) \right)$$

$$\leq n \cdot \frac{c}{m} = c\left(\frac{n}{m}\right) \leftarrow \text{load factor}$$

The expected length of a chain (# elements colliding with x)

$$= O\left(c \cdot \frac{n}{m}\right) = O(c) \text{ for } m \sim n$$

$= O\left(1 + \frac{n}{m}\right) = O(1)$ ...

for any subset $S \subset \mathcal{U}$

Does universal hash family exist?

Assume $N = |\mathcal{U}|$ is prime

Bertrand's lemma — Between any integer i and $2i$, there exists at least one prime

Consider $h_i(x) = (i + x) \bmod N \bmod m$

$0 \leq i \leq N-1$

$x \in \mathcal{U}$

$|H| = N$

Is this universal?

Is this universal?

Example 1 : Consider x, y where $y = x+m$

when is $h_i(x) = h_i(y)$: collision

i.e. for how may i's the above hold

$$h_i(x+m) = (x+i+m) \bmod N \bmod m$$

$$= (i+x) \bmod N \bmod m + m \bmod N \bmod m$$

$$= (i+x) \bmod N \bmod m$$

$$= h_i(x)$$

$\Rightarrow$ $h_i(x) = h_i(y)$ $\forall i$ and therefore this is $\boxed{\text{not a}}$ universal family

2^{nd} example

$$h'_i(x) = (i \cdot x \bmod N) \bmod m$$

How many solns exist for $x \neq y$

$$h'_i(x) = h'_i(y)$$

$$\Leftrightarrow \quad (i \cdot x \bmod N) \bmod m = (i \cdot y \bmod N) \bmod m$$

$$\Leftrightarrow \quad i \cdot (x-y) \bmod N = 0 \bmod m$$
$$= j \bmod N \text{ for}$$

$$\Leftrightarrow \quad i \equiv_N (x-y)^{-1} \cdot j \qquad j = 0, \pm m, \pm 2m, \ldots \pm \left\lfloor \frac{N}{m} \right\rfloor m$$

$(x-y)^{-1}$ is the inverse

of $(x-y) \neq 0$ in the multiplicative

prime field $\mathbb{Z}^*$ {

prime field $\mathbb{Z}_N = \{1, 2, 3, \ldots N-1\}$

So there are $\leq 2\left\lfloor \frac{N}{m} \right\rfloor$ solutions of i for which $h'_i(x) = h'_i(y)$, $x \neq y$

$|H| = N$ (or $N-1$ if we leave out $i=0$)

Check $\dfrac{2\left\lfloor \frac{N}{m} \right\rfloor}{N} \leq \dfrac{2 \cdot \frac{N}{m}}{N} = \dfrac{2}{m}$

which is 2-universal as per the definition

Cost of computing $h'(x)$

We must assume that any element

of $\mathcal{U}$ that consists of $\log_2 N$ bits can be operated on in $O(1)$ time (otherwise even the key can't be accessed in $O(1)$ time)

Also all operations like multiplication and divisions (mod function) can be performed in $O(1)$ time

So $h_i(x)$ can be computed in $O(1)$ time